

Independent Verification of the Blitz T-DCA BTCUSDT Perpetual Backtest

Preset N18/U75 (btc-futures-s14-40-n18-u75-20260824-public) | four cost scenarios | 2021-01-01 00:00 UTC to 2026-08-25 00:00 UTC (exclusive)

Author: Ranveer Verma (github.com/ranveer9) **Date:** 2026-09-29

Code version: v1.1.1 **Repository:** <https://github.com/ranveer9/blitz-btc-tdca-verification>

Commissioned by: Blitz Trading. The work was commissioned and paid for by Blitz Trading; the method, results and conclusions are the author's own. This is a historical verification of a backtest. It is not a guarantee of live trading performance.

Edition: PUBLIC EDITION - preset settings, sizing and cycle records omitted

Summary

Phase A - reproduction under the original assumptions: reproduced. An independent Python implementation written from the written specification, run on 2,062 checksum-verified Binance daily aggTrades files (3,198,919,867 trades, identical to the reference count), reproduces all four published cost scenarios. Final wallets, ROI, TP counts, liquidations, capital resets, maximum depth, fees, volume, funding and slippage agree within the stated tolerances (largest monetary difference 4.66e-10 USD). Annual figures, drawdown timestamps, all 14,018 baseline cycle open and close times and ladder depths, and 8,248 of 8,248 daily equity samples across the four scenarios also agree.

Phase B - adequacy of those assumptions. Reproducing the model does not validate it as a description of exchange execution. The model evaluates liquidation with trade prices and a fixed flat maintenance fraction rather than Binance mark prices and notional-tiered maintenance; it assumes every ladder order and every take-profit fills in full when a trade prints at or through its price; it applies the entry-fee rate to the immediate market-style first entry; it assumes instantaneous TP replacement and capital resets that the live bot does not perform automatically. Section 4 quantifies what the supplied data allow (fee sensitivities, latency-dependent cycles, shadow liquidation checks) and lists what remains unquantified. Zero modelled liquidations do not demonstrate exchange-level survival.

Added in v1.1.0. The client's closure-time MAE, capital-utilisation and liquidation-room statistics, defined after v1.0.0, were recomputed from their supplied definitions: 64 of 64 published values agree exactly (Section 3.8). Four quantified fill and latency sensitivities were added (Section 4.7): F1 one-tick trade-through: final wallet 2,249,550.34 USD (+2.20%), 14,158 TP cycles, MDD -86.30%; F2 volume at/through $\geq 1x$ size: final wallet 1,653,363.98 USD (-24.89%), 12,445 TP cycles, MDD -91.89%; F3 volume at/through $\geq 2x$ size: final wallet 1,620,936.43 USD (-26.36%), 12,311 TP cycles, MDD -92.81%; L1 1-second latency: final wallet 1,310,070.72 USD (-40.48%), 11,304 TP cycles, MDD -80.91%.

Publication boundary. At the client's written instruction, this public edition omits the exact ladder offsets, sizing weights, preset configuration values (including leverage, take-profit and reset settings, and fee and maintenance rates) and all trade/cycle-level records. The accompanying public code is a generic implementation without that configuration. **This public package therefore does not enable full public reproduction of the results.** The complete configuration, code and records were delivered privately to Blitz Trading. The omission does not limit any material finding: every finding is stated here.

1. Scope, inputs and data integrity

Frozen scope. BTCUSDT USD-M linear perpetual, long only; one published preset (N18/U75) evaluated under four cost scenarios: baseline (fees only), historical funding, funding plus 1 bp TP exit slippage, and funding plus 2 bp TP exit slippage. Window: 2021-01-01 00:00:00 UTC inclusive to 2026-08-25 00:00:00 UTC exclusive. Excluded: ETH and all other assets, the separately frozen 2026-08-25 to 2026-09-30 out-of-sample window (not downloaded, inspected or used), candidate or optimisation studies, and annual-reset variants.

Handover input used by this pipeline	SHA-256 vs client SHA256SUMS
presets.json	MATCH
btc_funding_rates.json	MATCH
s14_u75_instrumented.json	MATCH
btc-baseline-sparse-closed-cycles.csv	MATCH
s14_u75_equity_curves.json	MATCH
btc-public-risk-and-variants.json	MATCH
btc-reference-metadata.json	MATCH

Market data. 2,062 daily archives (2021-01-01 to 2026-08-24) were downloaded from data.binance.vision (futures/um/daily/aggTrades/BTCUSDT). 2,062 were verified against their official .CHECKSUM files at download time (files already present from an earlier verified download are logged as skipped); download log status: OK 2,062; no checksum failures or download errors occurred. The simulation processed 2,062 files and 3,198,919,867 trades; the reference run reports 3,198,919,867 (identical).

Funding. The supplied historical funding file (6,186 events, 2021-01-01 to 2026-08-24, hash-verified) was used as supplied. It was not re-downloaded; the client states no original download receipt exists. **Mark prices.** Binance publishes historical mark-price kline archives for USD-M futures; they were not part of the agreed inputs and were not used (see 4.1).

2. Method

Independence. The engine was written in Python 3.12 from the written specification and the supplied parameters only. The client's engine source code was not consulted. Prices and sizes are IEEE-754 binary64, timestamps are int64 milliseconds, and no tick or lot rounding is applied, as specified. At start-up the runner verifies that the engine constants equal the supplied configuration exactly and aborts otherwise. This applies to the Phase A engine and all Phase A results. The v1.1.0 risk-metric additions (Section 3.8) are different: they record the client's own statistics using the metric definitions and source excerpts of the client's instrumentation code, which the client supplied after v1.0.0. These were used only to reproduce those statistics and did not change the Phase A engine.

Event model reproduced (per aggTrade, in timestamp order; equal timestamps keep file order).

1. Settle every funding event due at or before this trade, at this trade's price (funding scenarios; skipped without payment while flat).
- 2a. If flat: open round 1 at the observed price and move to the next trade (no mark, TP or liquidation evaluation on the entry trade).
- 2b. If open: fill every ladder level whose price is at or above the observed price, each in full at its own level price (several levels can fill on one trade).
3. Mark equity (wallet plus unrealised PnL at the trade price); update drawdown statistics.
4. If the price is at or above the TP: close the whole cycle (a fixed take-profit target above the quantity-weighted average entry, with a fixed gross return on deployed margin), resize if the wallet reaches the reset growth threshold above the last reset level, and move to the next trade (no same-trade re-entry).
5. Otherwise evaluate the liquidation proxy: a fixed flat maintenance fraction on entry notional against the whole wallet, at the trade price; if breached, the wallet goes to zero and the run ends.

Accounting reproduced. Sizing: a fixed leverage, margins proportional to the ladder weights and U75 wallet usage, recomputed only at qualifying resets (never downward). Fees: the reference entry-fee rate (described in the specification as a maker assumption) on every entry, and the reference exit-fee rate charged on accumulated entry notional; entry fees accrue in the position and are deducted at the TP close. Exit slippage (scenarios 3-4) = quantity x TP price x slippage fraction. Volume counts entry notional of closed cycles once. Cashback is a reporting figure only. There are no annual resets. The final open position is not force-closed; the reported wallet is realised.

Tolerances (fixed before comparison). Monetary totals and annual values: absolute difference at most 1e-6 USD. ROI: at most 1e-9 percentage points. Counts and timestamps: exact. Maximum drawdown: the reference publishes four decimals, so ours rounded to four decimals must be identical. Daily equity samples: the reference stores two decimals, so the absolute difference must be at most 0.01 USD (exact two-decimal agreement is reported separately).

3. Phase A - reproduction under the original assumptions

3.1 Scenario totals

Baseline

Field	Independent	Reference	Difference	Result
Final realised wallet (USD)	2,201,183.984004	2,201,183.984004	4.66e-10	MATCH
ROI (%)	1550.892115	1550.892115	2.27e-13	MATCH
Closed TP cycles	14,018	14,018	0	MATCH
Liquidations	0	0	0	MATCH
Capital resets	29	29	0	MATCH
Max ladder depth	18	18	0	MATCH
Total fees (USD)	629,345.951653	629,345.951653	0	MATCH
Total volume (USD)	899,065,645.219029	899,065,645.219029	0	MATCH
Total funding (USD)	0.000000	0.000000	0	MATCH
Total slippage (USD)	0.000000	0.000000	0	MATCH
Account MTM max drawdown (%)	-81.3140	-81.3140	-2.84e-05	MATCH
aggTrades processed	3,198,919,867	3,198,919,867	0	MATCH
Max drawdown timestamp	2021-05-19 13:10:14.591 UTC	2021-05-19 13:10:14.591 UTC	0 ms	MATCH

Funding

Field	Independent	Reference	Difference	Result
Final realised wallet (USD)	1,801,697.354760	1,801,697.354760	0	MATCH
ROI (%)	1251.276394	1251.276394	0	MATCH
Closed TP cycles	14,018	14,018	0	MATCH
Liquidations	0	0	0	MATCH
Capital resets	27	27	0	MATCH
Max ladder depth	18	18	0	MATCH
Total fees (USD)	549,167.824954	549,167.824954	0	MATCH
Total volume (USD)	784,525,464.219271	784,525,464.219271	0	MATCH
Total funding (USD)	136,044.212944	136,044.212944	0	MATCH
Total slippage (USD)	0.000000	0.000000	0	MATCH
Account MTM max drawdown (%)	-78.1736	-78.1736	2.97e-05	MATCH
aggTrades processed	3,198,919,867	3,198,919,867	0	MATCH
Max drawdown timestamp	2021-05-19 13:10:14.591 UTC	2021-05-19 13:10:14.591 UTC	0 ms	MATCH

Funding + 1 bp

Field	Independent	Reference	Difference	Result
Final realised wallet (USD)	1,593,044.038240	1,593,044.038240	0	MATCH
ROI (%)	1094.786016	1094.786016	0	MATCH
Closed TP cycles	14,018	14,018	0	MATCH
Liquidations	0	0	0	MATCH
Capital resets	25	25	0	MATCH
Max ladder depth	18	18	0	MATCH
Total fees (USD)	503,793.353551	503,793.353551	0	MATCH
Total volume (USD)	719,704,790.787227	719,704,790.787227	0	MATCH
Total funding (USD)	123,423.590055	123,423.590055	0	MATCH
Total slippage (USD)	72,186.390516	72,186.390516	0	MATCH
Account MTM max drawdown (%)	-80.7259	-80.7259	2.62e-05	MATCH
aggTrades processed	3,198,919,867	3,198,919,867	0	MATCH
Max drawdown timestamp	2022-06-18 20:34:28.897 UTC	2022-06-18 20:34:28.897 UTC	0 ms	MATCH

Funding + 2 bp

Field	Independent	Reference	Difference	Result
Final realised wallet (USD)	1,410,768.973673	1,410,768.973673	-4.66e-10	MATCH
ROI (%)	958.079375	958.079375	-3.41e-13	MATCH
Closed TP cycles	14,018	14,018	0	MATCH
Liquidations	0	0	0	MATCH
Capital resets	24	24	0	MATCH
Max ladder depth	18	18	0	MATCH
Total fees (USD)	463,725.415220	463,725.415220	-5.82e-11	MATCH
Total volume (USD)	662,464,878.885767	662,464,878.885767	0	MATCH
Total funding (USD)	113,342.793060	113,342.793060	-7.28e-11	MATCH
Total slippage (USD)	132,890.454704	132,890.454704	0	MATCH
Account MTM max drawdown (%)	-81.6940	-81.6940	-1.16e-05	MATCH
aggTrades processed	3,198,919,867	3,198,919,867	0	MATCH
Max drawdown timestamp	2021-05-19 13:10:14.591 UTC	2021-05-19 13:10:14.591 UTC	0 ms	MATCH

3.2 Annual breakdown (continuous run; annual MDD resets only the peak statistic)

Each row compares eight annual fields with the client's instrumented reference: realised profit (excluding separately recorded funding), TP count, liquidations, fees, volume, funding, carried start wallet and intra-year MTM drawdown. 2026 covers 1 January to 24 August.

Baseline

Year	Profit (USD)	TP	Funding (USD)	Start wallet (USD)	MDD %	Max diff USD	Result
2021	325,700.33	6,026	0.00	133,333.00	-81.3140	1.16e-09	MATCH
2022	206,521.93	1,714	0.00	459,033.33	-75.7786	1.78e-09	MATCH
2023	350,792.46	2,325	0.00	665,555.26	-13.9638	2.98e-08	MATCH
2024	562,604.00	2,364	0.00	1,016,347.72	-28.7573	2.98e-08	MATCH
2025	401,518.07	1,096	0.00	1,578,951.72	-16.0344	2.98e-08	MATCH
2026	220,714.20	493	0.00	1,980,469.79	-64.5456	2.33e-10	MATCH

Funding

Year	Profit (USD)	TP	Funding (USD)	Start wallet (USD)	MDD %	Max diff USD	Result
2021	310,052.21	6,026	20,279.03	133,333.00	-78.1736	8.29e-10	MATCH
2022	186,384.06	1,714	17,137.27	423,106.18	-75.8780	1.09e-09	MATCH
2023	308,352.32	2,325	15,329.49	592,352.97	-14.3325	1.37e-09	MATCH
2024	480,160.58	2,364	37,613.36	885,375.80	-30.6906	2.79e-09	MATCH
2025	336,433.25	1,096	24,397.23	1,327,923.03	-17.5538	3.35e-10	MATCH
2026	183,026.14	493	21,287.83	1,639,959.04	-64.1132	7.28e-12	MATCH

Funding + 1 bp

Year	Profit (USD)	TP	Funding (USD)	Start wallet (USD)	MDD %	Max diff USD	Result
2021	287,763.06	6,026	19,887.14	133,333.00	-79.8654	3.86e-09	MATCH
2022	167,485.67	1,714	16,756.69	401,208.93	-80.7259	8.44e-10	MATCH
2023	272,198.53	2,325	13,980.22	551,937.90	-14.0080	3.61e-09	MATCH
2024	415,834.21	2,364	33,852.42	810,156.20	-30.7980	2.21e-09	MATCH
2025	286,364.04	1,096	21,546.06	1,192,138.00	-16.2899	2.04e-09	MATCH
2026	153,489.12	493	17,401.07	1,456,955.98	-65.5848	1.02e-10	MATCH

Funding + 2 bp

Year	Profit (USD)	TP	Funding (USD)	Start wallet (USD)	MDD %	Max diff USD	Result
2021	265,467.66	6,026	19,010.28	133,333.00	-81.6940	5.82e-10	MATCH
2022	150,035.53	1,714	15,329.13	379,790.38	-78.0581	1.49e-08	MATCH
2023	239,589.67	2,325	12,806.94	514,496.79	-13.7364	4.47e-08	MATCH
2024	360,155.51	2,364	30,710.46	741,279.51	-28.1725	4.66e-10	MATCH
2025	244,443.51	1,096	19,499.55	1,070,724.57	-16.7175	4.47e-08	MATCH
2026	131,086.88	493	15,986.43	1,295,668.52	-61.3145	1.49e-08	MATCH

3.3 Cycle-level comparison (baseline)

Item	Independent	Reference	Result
Closed cycles	14,018	14,018	MATCH
Cycle close timestamp equal (ms)	14,018	14,018	MATCH
Maximum ladder depth equal	14,018	14,018	MATCH
Liquidation flag equal	14,018	14,018	MATCH
Cycle open timestamp equal (ms)	14,018	14,018	MATCH
Zero-duration cycles	1,025	1,025	MATCH

All cycle open timestamps are also identical.

Depth bucket (TP cycles)	Independent	Reference	Result
1-5	13,295	13,295	MATCH
6-10	598	598	MATCH
11-15	113	113	MATCH
16-18	12	12	MATCH

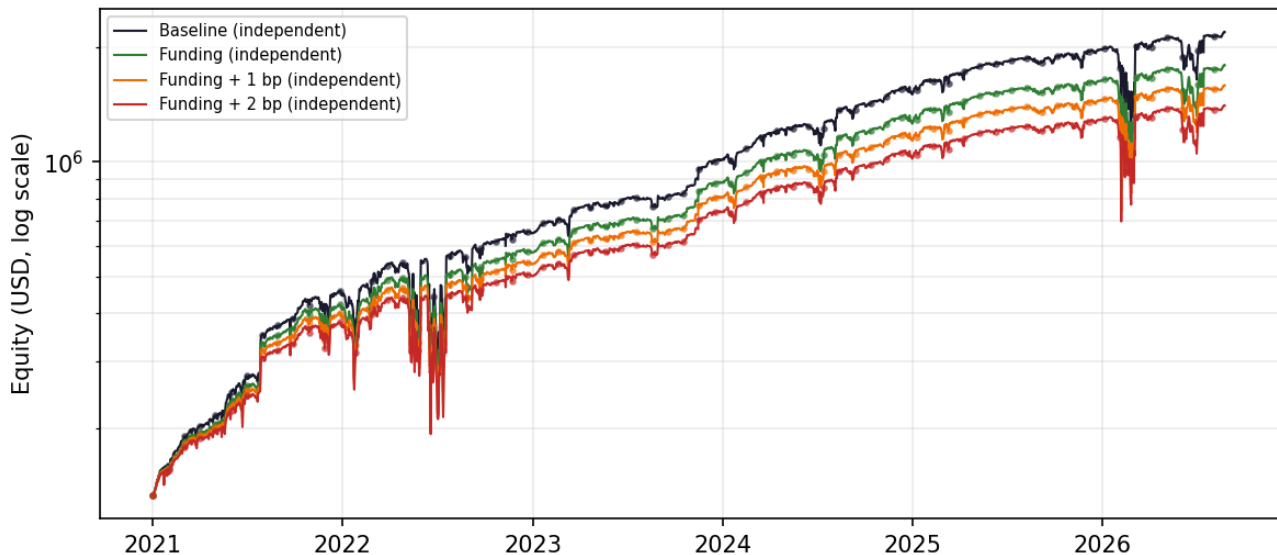
Cycle timing is the same in every cost scenario: Funding 14,018 of 14,018 cycles have the baseline close time and depth; Funding + 1 bp 14,018 of 14,018 cycles have the baseline close time and depth; Funding + 2 bp 14,018 of 14,018 cycles have the baseline close time and depth. This is expected: the average entry and TP depend only on prices and the ladder shape, not on the wallet size. Costs change position sizes and wallet, not when cycles open or close.

3.4 Daily sampled equity curves

The reference stores the first marked equity observation of each UTC day (not a close series). This implementation records the same observation with the same timing rule.

Scenario	Samples (ind./ref.)	Timestamps equal	Exact at 2 dp	Within 0.01 USD	Max diff USD	Result
Baseline	2,062 / 2,062	2,062	2,062	2,062	0.00	MATCH
Funding	2,062 / 2,062	2,062	2,062	2,062	0.00	MATCH
Funding + 1 bp	2,062 / 2,062	2,062	2,062	2,062	0.00	MATCH
Funding + 2 bp	2,062 / 2,062	2,062	2,062	2,062	0.00	MATCH

Daily sampled MTM equity: independent run (lines) vs client reference (dots, every 30th)



3.5 Internal consistency of the independent cycle ledgers

Check	Baseline	Funding	Funding + 1 bp	Funding + 2 bp
Ledger rows = TP cycles + liquidations	PASS	PASS	PASS	PASS
Sum of cycle fees = total fees	PASS	PASS	PASS	PASS
Sum of cycle entry notional = total volume	PASS	PASS	PASS	PASS
Sum of cycle slippage = total slippage	PASS	PASS	PASS	PASS
Cycle funding + open-position funding = total funding	PASS	PASS	PASS	PASS
Wallet chain: before - funding + net = after, every cycle	PASS	PASS	PASS	PASS
Wallet chain reproduces the final wallet	PASS	PASS	PASS	PASS

Check	Baseline	Funding	Funding + 1 bp	Funding + 2 bp
Per-cycle gross profit, exit fee and TP follow the stated formulas	PASS	PASS	PASS	PASS
Deepest cycle = reported maximum depth	PASS	PASS	PASS	PASS

Sum checks allow for floating-point summation order (largest differences: Baseline fees 1.47e-08, funding 0; Funding fees -2.40e-08, funding 0; Funding + 1 bp fees -4.31e-09, funding 0; Funding + 2 bp fees 6.69e-09, funding 0 USD).

3.6 Terminal open position and economically net terminal equity

The run ends with an open position, which is not force-closed. The reported wallet is closed-cycle realised wallet. The separately labelled net terminal equity below adds the unrealised PnL at the last trade and deducts the accrued (not yet charged) entry fees, the estimated exit fee (reference convention: on entry notional) and, in the slippage scenarios, the estimated exit slippage at the last price. Funding already paid on the open position is inside the wallet.

Item	Baseline	Funding	Funding + 1 bp	Funding + 2 bp
Status	OPEN	OPEN	OPEN	OPEN
Opened	2026-08-24 16:17:30.514 UTC	2026-08-24 16:17:30.514 UTC	2026-08-24 16:17:30.514 UTC	2026-08-24 16:17:30.514 UTC
Ladder round reached	4	4	4	4
Average entry	79,208.4350	79,208.4350	79,208.4350	79,208.4350
Take-profit price	79,446.0603	79,446.0603	79,446.0603	79,446.0603
Last trade price	78,953.00	78,953.00	78,953.00	78,953.00
Realised wallet	2,201,183.98	1,801,697.35	1,593,044.04	1,410,768.97
Gross unrealised PnL	-1,160.46	-950.89	-785.17	-717.04
Economically net terminal equity	2,199,771.62	1,800,540.06	1,592,064.17	1,409,851.96

Baseline item	Independent	Reference	Result
Opened (ms)	2026-08-24 16:17:30.514 UTC	2026-08-24 16:17:30.514 UTC	MATCH
Round	4	4	MATCH
Average entry	79,208.434957	79,208.434957	MATCH
Take-profit	79,446.060262	79,446.060262	MATCH
Last trade price	78,953.000000	78,953.000000	MATCH
Gross unrealised PnL	-1,160.463473	-1,160.463473	MATCH
MTM after accrued and exit fees	2,199,771.624911	2,199,771.624911	MATCH

3.7 Boundary-case checks

Twelve targeted checks drive the engine with small synthetic trade sequences and compare the outcome with the specification (12 of 12 passed). They cover the cases the engagement named explicitly: several ladder levels crossed by one trade (C02), a TP update after an additional fill (C03), and the order of funding versus the liquidation check (C07).

ID	Check	Result
C01	Flat entry trade	PASS
C02	Multiple ladder levels crossed at one event	PASS
C03	TP update after an additional fill	PASS
C04	No TP overshoot price improvement	PASS
C05	No same-event re-entry after TP	PASS
C06	Fee timing, exit-fee basis, slippage, volume	PASS
C07	Funding applied before liquidation check	PASS
C08	Funding while flat; multiple pending events	PASS
C09	Liquidation outcome (wallet 0, terminate)	PASS
C10	Capital reset threshold; no downward resize	PASS
C11	Liquidation checked after full DCA batch	PASS
C12	Timestamp ordering and window filter	PASS

3.8 Client risk metrics (closure-time definitions, added in v1.1.0)

The client supplied the exact definitions and source excerpts behind its published MAE, capital-utilisation and liquidation-room figures after v1.0.0. They are recorded by a direct port of that source inside the compiled engine (4.7), whose four reference simulations reproduce the Phase A runs bit-for-bit on the full data; the recorder was checked cycle by cycle against an independent Python port. Per closed cycle, at the closure callback: U = the most negative (price x quantity - cost) over the cycle's marks, with its price P^* ; MAE vs account = $100 U / W$; MAE vs capital = $100 U / M$; utilisation = $100 M / W$; liquidation room = $100 (L / P^* - 1)$, with L the proxy liquidation price from the closing position and W . M is the cycle's allocated margin and W the wallet at closure after the cycle's realised PnL. Only closed cycles count; summaries are unweighted, quantiles use the client's no-interpolation index rule, values are rounded to four decimals and the raw sign is kept.

Interpretation. Utilisation is one closure-time ratio per cycle, not a measured intra-cycle peak or a time-weighted mean. MAE is normalised at closure; it is not a running unrealised-PnL/wallet minimum and not the account drawdown. The liquidation-room statistic combines the worst-PnL price with the closing position and wallet, so it is not a same-timestamp distance; this report's contemporaneous minimum (4.1) is a different quantity and is kept. Neither establishes mark-price or tiered-maintenance safety.

Baseline

Statistic	Independent	Reference	Result
MAE vs account % mean	-0.1499	-0.1499	MATCH
MAE vs account % median	-0.0121	-0.0121	MATCH
MAE vs account % p5	-0.3504	-0.3504	MATCH
MAE vs account % p1	-2.0436	-2.0436	MATCH
MAE vs account % worst	-80.6409	-80.6409	MATCH
MAE vs capital % mean	-2.5122	-2.5122	MATCH
MAE vs capital % median	-1.1214	-1.1214	MATCH
MAE vs capital % p5	-8.4939	-8.4939	MATCH
MAE vs capital % p1	-28.5121	-28.5121	MATCH
MAE vs capital % worst	-112.677	-112.677	MATCH
Utilisation % mean	1.7388	1.7388	MATCH
Utilisation % median	1.5273	1.5273	MATCH
Utilisation % p95	4.195	4.195	MATCH
Utilisation % max	71.9167	71.9167	MATCH
Liquidation room % signed max (key 'min')	-6.3383	-6.3383	MATCH
Liquidation room % count with liq. line	21	21	MATCH

Funding

Statistic	Independent	Reference	Result
MAE vs account % mean	-0.1505	-0.1505	MATCH
MAE vs account % median	-0.0121	-0.0121	MATCH
MAE vs account % p5	-0.3531	-0.3531	MATCH
MAE vs account % p1	-2.0934	-2.0934	MATCH
MAE vs account % worst	-77.5079	-77.5079	MATCH
MAE vs capital % mean	-2.5122	-2.5122	MATCH
MAE vs capital % median	-1.1214	-1.1214	MATCH
MAE vs capital % p5	-8.4939	-8.4939	MATCH
MAE vs capital % p1	-28.5121	-28.5121	MATCH
MAE vs capital % worst	-112.677	-112.677	MATCH
Utilisation % mean	1.7391	1.7391	MATCH
Utilisation % median	1.5272	1.5272	MATCH
Utilisation % p95	4.1849	4.1849	MATCH
Utilisation % max	74.1574	74.1574	MATCH
Liquidation room % signed max (key 'min')	-7.7966	-7.7966	MATCH
Liquidation room % count with liq. line	21	21	MATCH

Funding + 1 bp

Statistic	Independent	Reference	Result
MAE vs account % mean	-0.151	-0.151	MATCH
MAE vs account % median	-0.0121	-0.0121	MATCH
MAE vs account % p5	-0.3498	-0.3498	MATCH
MAE vs account % p1	-2.0257	-2.0257	MATCH
MAE vs account % worst	-81.1378	-81.1378	MATCH
MAE vs capital % mean	-2.5122	-2.5122	MATCH
MAE vs capital % median	-1.1214	-1.1214	MATCH
MAE vs capital % p5	-8.4939	-8.4939	MATCH
MAE vs capital % p1	-28.5121	-28.5121	MATCH
MAE vs capital % worst	-112.677	-112.677	MATCH
Utilisation % mean	1.7386	1.7386	MATCH
Utilisation % median	1.5275	1.5275	MATCH
Utilisation % p95	4.1941	4.1941	MATCH
Utilisation % max	73.754	73.754	MATCH
Liquidation room % signed max (key 'min')	-5.9665	-5.9665	MATCH
Liquidation room % count with liq. line	21	21	MATCH

Funding + 2 bp

Statistic	Independent	Reference	Result
MAE vs account % mean	-0.1498	-0.1498	MATCH
MAE vs account % median	-0.0121	-0.0121	MATCH
MAE vs account % p5	-0.3518	-0.3518	MATCH
MAE vs account % p1	-2.0908	-2.0908	MATCH
MAE vs account % worst	-81.0341	-81.0341	MATCH
MAE vs capital % mean	-2.5122	-2.5122	MATCH
MAE vs capital % median	-1.1214	-1.1214	MATCH
MAE vs capital % p5	-8.4939	-8.4939	MATCH
MAE vs capital % p1	-28.5121	-28.5121	MATCH
MAE vs capital % worst	-112.677	-112.677	MATCH
Utilisation % mean	1.7385	1.7385	MATCH
Utilisation % median	1.5275	1.5275	MATCH
Utilisation % p95	4.1974	4.1974	MATCH
Utilisation % max	71.9171	71.9171	MATCH
Liquidation room % signed max (key 'min')	-6.1633	-6.1633	MATCH
Liquidation room % count with liq. line	21	21	MATCH

3.9 Phase A conclusion

Under the original assumptions, the specified strategy and accounting reproduce the published BTC results for all four cost scenarios, at total, annual, cycle and daily-equity level, within the tolerances fixed in Section 2. No difference required explanation. This establishes that the published figures follow from the written rules and the public data. It does not establish that those rules describe achievable exchange execution (Section 4).

4. Phase B - adequacy of the assumptions

This section assesses what the reference assumptions can and cannot support. Nothing here altered the Phase A runs. Figures computed under changed assumptions are labelled as sensitivities; they are first-order estimates on the reference path unless stated otherwise, because a changed cost changes wallet, sizing and compounding thereafter.

4.1 Liquidation treatment

What the model does. Price source: the aggTrade (last) price. Time resolution: every trade, millisecond timestamps, evaluated once after the full same-trade ladder batch and after funding. Maintenance: a fixed flat maintenance fraction of entry notional against the whole wallet (cross-wallet proxy). There are no exchange notional tiers, no liquidation fee, no mark or index premium, no partial liquidation and no ADL. Accrued entry fees are not deducted from the wallet until the TP, so the wallet used in the check is slightly overstated during open cycles (at most 0.0725% of wallet in the baseline run; immaterial).

Last price is not inherently a conservative substitute for mark price. Binance triggers liquidation on the mark price, a separately computed series derived from index prices and a premium component. During dislocations the last price can trade below the mark price (the model then liquidates too early) or the mark price can fall below the last price or move while trades are sparse (the model then liquidates too late or not at all). The direction is not known a priori. Historical mark-price klines are publicly archived at 1-minute resolution but were not an agreed input. **The mark-price effect on liquidation is not quantified in this report.**

How close the modelled path came. A liquidation price existed (the wallet could not absorb a fall to zero) during 21 baseline cycles. The smallest distance between a trade price and the modelled liquidation price was 6.0414% of price (2021-05-19T13:10:14.591Z). The number of cycles with a liquidation line agrees with the client's instrumented output (21). The client also publishes a liquidation-room percentage (6.3383% for this scenario) whose formula and measurement point were supplied after v1.0.0; it is recomputed and compared in Section 3.8. The figure above is this report's own metric: the minimum over all trades of (trade price - proxy liquidation price) / trade price.

Shadow liquidation check (hypothetical, does not alter the run)	Baseline	Funding	Funding + 1 bp	Funding + 2 bp
Intermediate check after each individual level of a multi-level fill	none	none	none	none
Flat maintenance 1.0% instead of the reference fraction: first breach	none	none	none	none
Flat maintenance 2.5% instead of the reference fraction: first breach	none	none	none	none
Smallest distance to liquidation at 1.0% / 2.5%	5.40% / 3.46%	6.85% / 4.92%	5.50% / 3.58%	5.25% / 3.31%

The 1.0% and 2.5% cases are flat hypothetical rates chosen for sensitivity, not Binance's historical tier schedule, which raises the maintenance rate as position notional grows and changed over 2021-2026. The tier schedule was not supplied and was not reconstructed. A breach means the modelled path would have ended at that point under that assumption; later results would not exist. Zero liquidations under the reference proxy do not prove exchange-level survival.

4.2 Fills, liquidity and latency

A one-tick trade-through does not guarantee queue clearance or a full fill. The reference fills each ladder level in full when any trade prints at or through its price, and closes the whole position when any trade prints at or above the TP. It uses neither the aggTrade quantity nor the order book, queue position, partial fills or order-entry latency. A print at the limit price shows that some quantity traded there, not that a resting order of the modelled size would have been filled. The fill rule is therefore a modelling assumption, not evidence of executable quantity; its effect under explicit tick and volume rules is quantified in 4.7.

Gaps. In the baseline run 594 trades crossed two or more levels at once (at most 13 levels in one trade); all were filled at their level prices. Resting limit orders would fill at their limit prices, so this is consistent if all levels are resting orders; if levels are placed as they are reached, gapped levels would fill later or at worse prices.

TP replacement latency. After every additional fill the TP moves down to the new average entry; the model re-prices it instantaneously. A live bot must cancel and replace an exchange order. The model also opens a new cycle on the very next trade after a TP. 1,025 baseline cycles opened and closed within the same millisecond and 2,937 within one second; together the sub-second cycles contributed 313,429.62 USD of realised profit before funding (15.16% of the closed-cycle total). These outcomes depend on zero-latency order placement and are not demonstrably achievable. The effect of real latency on the compounded path is quantified for a 1-second delay in 4.7.

4.3 Fees

The model charges the reference entry-fee rate (described in the specification as a maker assumption) on every entry, and the reference exit-fee rate charged on accumulated entry notional. Two assumptions pull in opposite directions:

First entry. Round 1 opens at the next traded price immediately after a TP, which in practice is a marketable (taker) order.

Charging an illustrative taker rate of 0.04% or 0.05% on round 1 instead would add 83,912.95 or 125,869.43 USD of fees over the baseline path (4.06% or 6.09% of closed-cycle net profit).

Exit. If TP exits execute as resting maker orders at 0.02% (or taker at 0.04%), the reference exit fee overstates cost by 269,719.69 (or 89,906.56) USD (13.04% or 4.35% of closed-cycle net profit). Charging the exit fee on exit turnover instead of entry notional would add only 1,348.60 USD.

The net direction depends on the order types the live bot uses, which were outside the supplied material. These are first-order, non-compounded figures on the reference path; account VIP tiers, fee discounts and historical fee-schedule changes are not modelled.

4.4 Funding

Funding is settled at the first trade at or after each event, using that trade's price as a mark-price proxy; events while flat are skipped. Binance settles on the mark price at the funding time. Historical funding reduced the final realised wallet by 18.15% versus the baseline (total funding 136,044.21 USD), through lower compounding, not just the payments. The difference between the trade-price proxy and the exact mark price at settlement is not quantified; it is expected to be small relative to the funding amounts themselves.

4.5 Slippage

Slippage is applied only to TP exits, uniformly per unit of price, with no entry slippage and no dependence on size, volatility or liquidity. 1 bp and 2 bp lower the final realised wallet by 27.63% and 35.91% versus the baseline (with funding). The strategy's per-cycle margin is small, so results are highly sensitive to per-cycle costs. Slippage during fast markets, when deep ladder levels fill, can exceed 2 bp; that is not quantified.

4.6 Capital resets, drawdown and holding time

The model resizes the ladder automatically when the wallet reaches the reset growth threshold above the last reset level (29 resets in the baseline). The specification states that the live bot does not do this automatically and users must reset manually; the compounded figures assume resets at the modelled times. Account MTM drawdown reached -81.31% (2021-05-19 13:10:14.591 UTC); intra-year MTM drawdown exceeded 60% in 2021, 2022, 2026. Cycle holding time: median 2.6 min, mean 3.53 h, 95th percentile 8.5 h, longest 1463 h (2026-05-14 to 2026-07-14, depth 17). 12 cycles reached depth 16 or more, 6 the full ladder.

The client's MAE, capital-utilisation and liquidation-room statistics are recomputed and compared in Section 3.8.

4.7 Quantified fill and latency sensitivities (added in v1.1.0)

Version 1.0.0 documented the fill, queue and latency limitations without testing them. v1.1.0 adds four explicit alternative execution rules. Each is run on the baseline cost scenario so the execution effect is isolated, with every other rule unchanged. Round 1 still opens at the observed price. They are Phase B sensitivities under changed assumptions, not corrections of Phase A. They were run with a compiled implementation of the same event model that, with the alternative rules switched off, reproduces all four Phase A runs bit-for-bit on the full data (scalars, all cycles and all daily equity samples), and passed 14 of 14 targeted rule checks.

Rule	Definition
F1	A ladder level fills only if a trade prints at least one tick (0.10 USD) below it; the TP only if a trade prints at least one tick above it.
F2	An order fills only once the cumulative aggTrade quantity printed at or through its price since it became active reaches its own size (no queue ahead). Ladder levels are active from cycle entry; the TP from its last re-pricing. All-or-nothing fills.
F3	As F2, with a queue ahead equal to the order's own size (2x its size must print).
L1	A new cycle opens only on the first trade at least 1 s after the TP; ladder and TP become active 1 s after entry; after each additional fill the TP is re-placed and active again 1 s later.

Run (baseline costs)	Final wallet (USD)	vs reproduced	TP cycles	Liquidations	Max DD %	Liquidated on
Reference rules (Phase A)	2,201,183.98	0.00%	14,018	0	-81.31	-
F1 one-tick trade-through	2,249,550.34	+2.20%	14,158	0	-86.30	-
F2 volume at/through >= 1x size	1,653,363.98	-24.89%	12,445	0	-91.89	-
F3 volume at/through >= 2x size	1,620,936.43	-26.36%	12,311	0	-92.81	-
L1 1-second latency	1,310,070.72	-40.48%	11,304	0	-80.91	-

Limits of these tests. aggTrade quantity is used as a proxy for the volume that would have reached a resting order; the true queue ahead is unknown because order-book depth is not in the data, so F2 and F3 bracket rather than measure queue effects. Partial fills are not modelled (all-or-nothing). The tick size is taken as 0.10 USD for the whole window; historical tick-size changes were not reconstructed. L1 tests one latency value. A liquidation in a sensitivity run ends that run; results after that date do not exist under that rule. A higher result under a stricter rule is possible because skipped or delayed fills change position sizes and cycle timing along the path; it is not evidence that the stricter execution would be favourable in live trading.

5. Findings

5.1 Reproduction (Phase A). Reproduced. The published BTC N18/U75 results for all four cost scenarios follow from the written specification and public Binance data, independently implemented, at total, annual, cycle and daily-equity level. With the definitions supplied after v1.0.0, 64 of 64 published closure-time risk statistics are reproduced exactly (3.8).

5.2 Negative findings and material limitations (Phase B).

1. Liquidation is modelled on trade prices with a flat maintenance fraction. Last price is not a conservative substitute for mark price, and notional-tiered maintenance is not modelled. Zero liquidations under the proxy do not demonstrate survival at the exchange.
2. The path came within 6.04% of the modelled liquidation price. Under a hypothetical flat 2.5% maintenance fraction the closest approach is 3.46%, still without a breach; neither result reflects mark prices or Binance's tiers (4.1).
3. Fills are assumed complete on any print at or through the order price. A trade-through does not guarantee queue clearance or a full fill; partial fills, queue position and liquidity are not modelled.
4. 2,937 cycles lasted under one second and rely on zero-latency order placement and instantaneous TP replacement.
5. The immediate first entry is charged the entry-fee rate that the specification describes as a maker assumption, although it is taker-like; the exit fee may overstate cost if exits are maker. The net fee effect depends on live order types (4.3).
6. Results are highly sensitive to per-cycle costs: funding, 1 bp and 2 bp exit slippage reduce the final wallet by 18.1%, 27.6% and 35.9%.
7. Account drawdown reached -81.3%; compounding assumes capital resets that the live bot does not perform automatically.
8. Quantified execution sensitivities on the baseline path (4.7): F1 one-tick trade-through: final wallet 2,249,550.34 USD (+2.20%), 14,158 TP cycles, MDD -86.30%; F2 volume at/through $\geq 1x$ size: final wallet 1,653,363.98 USD (-24.89%), 12,445 TP cycles, MDD -91.89%; F3 volume at/through $\geq 2x$ size: final wallet 1,620,936.43 USD (-26.36%), 12,311 TP cycles, MDD -92.81%; L1 1-second latency: final wallet 1,310,070.72 USD (-40.48%), 11,304 TP cycles, MDD -80.91%. They use aggTrade quantity as a volume proxy with an assumed queue and all-or-nothing fills.

5.3 Unquantified effects and unavailable inputs.

- Mark-price versus trade-price liquidation timing (public 1-minute mark-price archives exist but were not an agreed input).
- Binance historical maintenance-margin tier schedules, liquidation fees and ADL (not supplied).
- True queue position beyond the tested 1x and 2x volume rules, partial fills and order-book depth.
- Latency values other than the tested 1 second; order rejections; exchange minimums and lot rounding.
- The live bot's actual order types (maker/taker) and account fee tier.
- Exact mark prices at funding settlement (the supplied funding file has no original download receipt).
- Size- and volatility-dependent slippage beyond the uniform 1-2 bp exit scenarios; entry slippage.

6. Reproduction, data retrieval and code version

Data. Daily archives:

<https://data.binance.vision/data/futures/um/daily/aggTrades/BTCUSDT/BTCUSDT-aggTrades-YYYY-MM-DD.zip> with the matching .zip.CHECKSUM, for 2021-01-01 to 2026-08-24 (2,062 files). src/downloader.py reads the client's download manifest, verifies each archive against its official checksum and writes logs/download_log.csv.

Run. pip install -r requirements.txt; python src/downloader.py ...; then, in four terminals, python run_scenario.py baseline | funding_only | funding_slip1bp | funding_slip2bp (about 5 to 6 hours each on a desktop CPU, run in parallel); python tests/boundary_checks.py; python tests/variant_checks.py; python run_fast.py (compiled engine, one pass over the data running all eight v1.1.0 simulations, about 1 hour); python finalize.py. Python 3.12; numpy 2.1.3; pandas 2.2.3; reportlab 4.2.5; matplotlib 3.9.3; requests 2.32.3; tqdm 4.67.1; numba 0.61.2.

Code version. v1.1.1 at <https://github.com/ranveer9/blitz-btc-tdca-verification>. The public repository contains generic code that requires the private configuration file; without it the results cannot be reproduced.

Changes in v1.1.0

Added Section 3.8 (client closure-time MAE, utilisation and liquidation-room statistics reproduced from definitions supplied after v1.0.0) and Section 4.7 (four quantified fill and latency sensitivities with a validated compiled engine); updated 4.1, 4.2, 4.6 and Section 5 accordingly. Phase A results, all v1.0.0 findings and negative findings are unchanged.

Changes in v1.1.1

Wording clarifications only, at the client's request: Section 2 distinguishes the independently implemented Phase A engine from the v1.1.0 risk-metric additions, which used client-supplied definitions and source excerpts; the README states that liquidity and latency are not modelled in the reference assumptions and points to the v1.1.0 sensitivity tests. No simulation was rerun; all results, findings and limitations are

unchanged.

7. Publication and attribution

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